NUS GRIP
Ambient Systems

Assignment #4

October 6, 2020

Assignments 1-3 Revisited

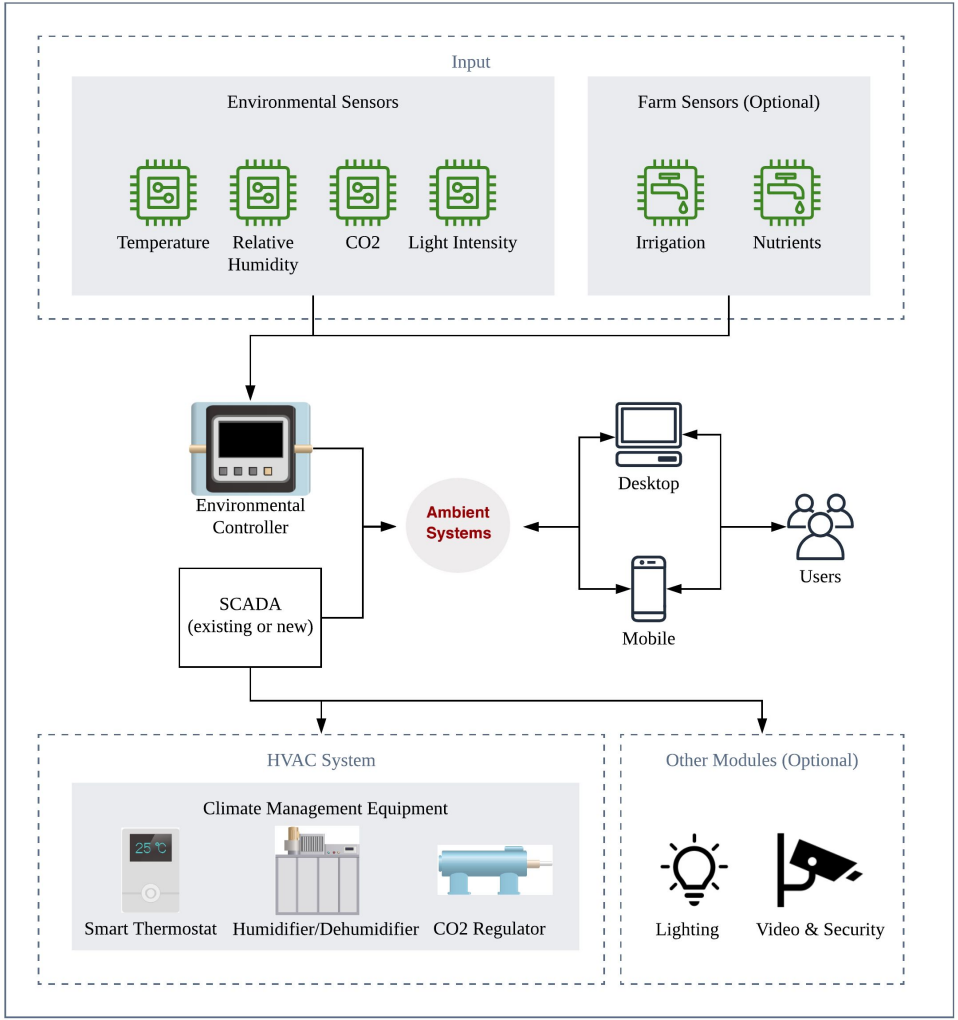


MVP: 3-part solution



IoT Sensor Network	Software Platform	Service Model
• Temperature sensor (0 to 50 C +/- 0.1 C) • Relative humidity sensor (0% to 100% +/- 1%) • CO2 sensor (ppm +/- 5 ppm) • Light intensity sensor • Relevant connectors, mountings, gateway, cabling, and controls	• Real-time data delivered using modern user interface design • Provides alerts via email or text when environmental conditions are beyond optimal range • Provides recommendations to remedy the situation	• Provide a “Climate-as-a-Service” • User pays an annual subscription fee based on farm type and area • Streamline financing of the CAPEX and OPEX of the system

Ambient MVP High-level Schematic



Pricing: TEV and PEV

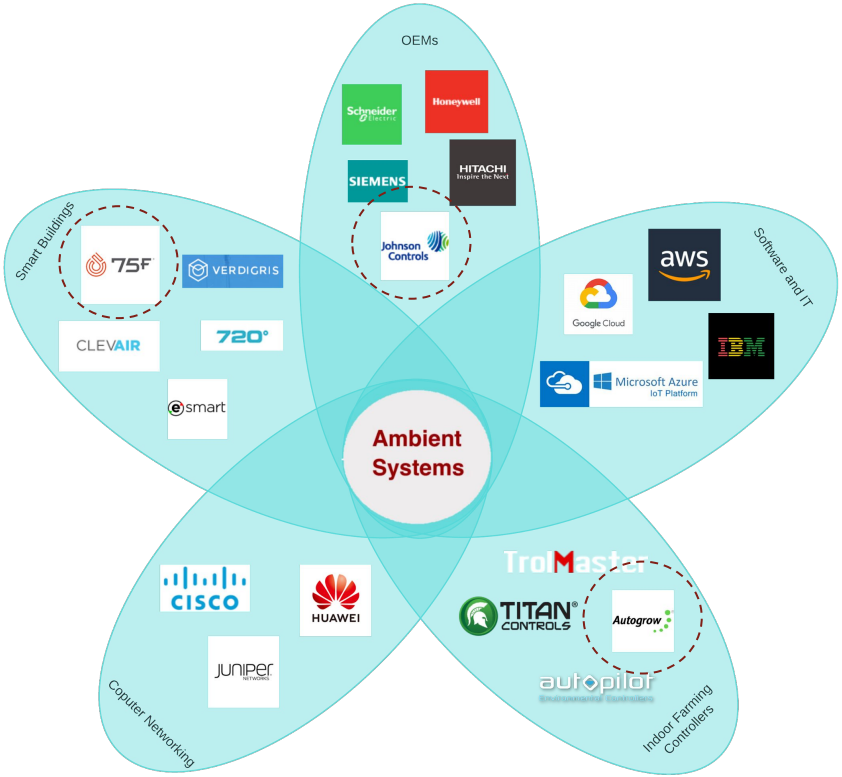
Value Stream	Value Component*	Best Alternative	Ambient
Implementation Value	Hardware	$3,000.00	$2,325.00
	Installation (24 hours * $150/hr technician cost)	$3,600.00	$3,600.00
Operational Value	Data Management (1 day / month * 36 months product lifetime * $25/hr personnel cost)	$0.00	$7,200.00
	System Maintenance (10% of Implementation Value * 5 years)	$0.00	$1,777.50
Use Case Specific Value	Higher Yield	$0.00	TBD
	More Accurate Harvest Time Prediction	$0.00	TBD
	Higher Quality	$0.00	TBD
	Total Economic Value (TEV)	**$6,600.00**	**$14,902.50**
	TEV per m2 of farming area	**$221.90**	**$501.03**

*Main Assumptions: product lifetime and subscription contract of 3 years (36 months); 1 farming unit covers 30m2.

Section 1: Competitive Analysis



Competitive Analysis: Petal Diagram



Ambient is positioned at the intersection of multiple segments in order to provide a modern building automation system (BAS) that is tailored for the indoor farming market.

Competitive Benchmark

	Ambient	Autogrow	75F	Johnson Controls
Segment	Multiple	Indoor Farming Controller	Smart Buildings	OEM
Proximity to Customer	✓ Singapore	✗ US	✗ US	✓ International
Specific vs. Generic	✓ Indoor Farms	✓ Indoor Farms	✗ No	✗ No
Modern Web-based UI	✓ Yes	✗ No	✓ Yes	✗ Some
Wireless IoT Network	✓ Yes	✗ No	✓ Yes	✓ Yes
Advanced Optimization Methods	✓ ML models	✗ No	✗ Some	✗ Some
Pricing	Quotation (based on area)	$3000 for one unit + Installation Fees	Quotation	Quotation

Pricing Benchmark

Value Stream	Value Component	Price per 1 unit (30 m2)	Price for 10 units (300 m2)	Price for 200 units (6000 m2)
Implementation Value	Hardware	$4,000.00	$40,000.00	$800,000.00
	Installation	$3,600.00	$36,000.00	$720,000.00
Operational Value	Data Management	$0.00	$0.00	$0.00
	System Maintenance	$760.00	$7,600.00	$152,000.00
Use Case Specific Value	Higher Yield	$0.00	$0.00	$0.00
	More Accurate Harvest Time Prediction	$0.00	$0.00	$0.00
	Higher Quality	$0.00	$0.00	$0.00
	Estimated Total Price	**$8,360.00**	**$83,600.00**	**$1,672,000.00**

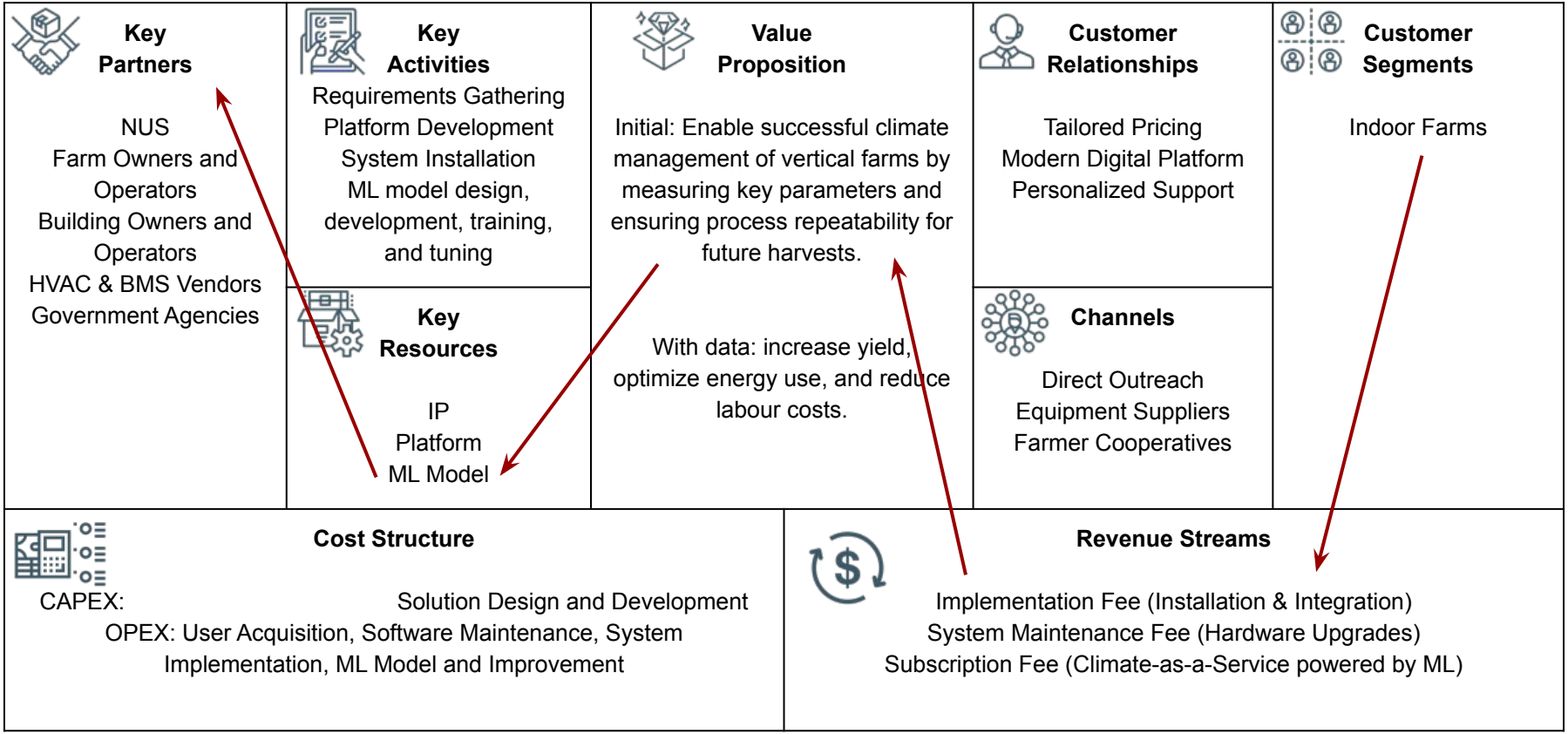
Pricing benchmark based on a customer budget and planned number of installations.

Section 2: Business Model Canvas



Value Proposition Canvas: Vertical Farms

Business Model Canvas



Section 3: Financial Projections



Sensitivity Analysis: Scenario Planning

Parameter	Worst	Baseline	Best
COGS	$90/m2	$80/m2	$70/m2
Sale Price	$400/m2	$500/m2	$600/m2
Units Sold Growth Rate	1.5X (50% growth annually)	2X (double annually)	3X (triple annually)
	$877k	$230k	$120k
Valuation	-$710k	$6.5M	$47.4M

Financial projections are most sensitive to units sold and annual sales growth rate.

Financial Projections: Worst

Inputs

Product price ($/m^2)	$400.00
Product COGS ($/m^2)	$90.00
Discount Rate	5.00%

	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	$400,000.00	$600,000.00	$900,000.00	$1,350,000.00	$2,025,000.00
Area under management (m2)	1000	1500	2250	3375	5062.5
minus Cost of Goods Sold	$90,000.00	$135,000.00	$202,500.00	$303,750.00	$455,625.00
Gross Margin	$310,000.00	$465,000.00	$697,500.00	$1,046,250.00	$1,569,375.00
Expenses	$650,000.00	$800,000.00	$900,000.00	$1,000,000.00	$1,500,000.00
Profit & Loss = Net Cash Flow	-$340,000.00	-$335,000.00	-$202,500.00	$46,250.00	$69,375.00

	-$340,000.00	-$675,000.00	-$877,500.00	-$831,250.00	-$761,875.00

Valuation (NPV)	-$710,184.40

Financial Projections: Baseline

Inputs

Product price ($/m^2)	$500.00
Product COGS ($/m^2)	$80.00
Discount Rate	5.00%

	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	$500,000.00	$1,000,000.00	$2,000,000.00	$4,000,000.00	$8,000,000.00
Area under management (m2)	1000	2000	4000	8000	16000
minus Cost of Goods Sold	$80,000.00	$160,000.00	$320,000.00	$640,000.00	$1,280,000.00
Gross Margin	$420,000.00	$840,000.00	$1,680,000.00	$3,360,000.00	$6,720,000.00
Expenses	$650,000.00	$800,000.00	$900,000.00	$1,000,000.00	$1,500,000.00
Profit & Loss = Net Cash Flow	-$230,000.00	$40,000.00	$780,000.00	$2,360,000.00	$5,220,000.00

	-$230,000.00	-$190,000.00	$590,000.00	$2,950,000.00	$8,170,000.00

Valuation (NPV)	$6,522,611.32

Financial Projections: Best

Inputs

Product price ($/m^2)	$600.00
Product COGS ($/m^2)	$70.00
Discount Rate	5.00%

	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	$600,000.00	$1,800,000.00	$5,400,000.00	$16,200,000.00	$48,600,000.00
Area under management (m2)	1000	3000	9000	27000	81000
minus Cost of Goods Sold	$70,000.00	$210,000.00	$630,000.00	$1,890,000.00	$5,670,000.00
Gross Margin	$530,000.00	$1,590,000.00	$4,770,000.00	$14,310,000.00	$42,930,000.00
Expenses	$650,000.00	$800,000.00	$900,000.00	$1,000,000.00	$1,500,000.00
Profit & Loss = Net Cash Flow	-$120,000.00	$790,000.00	$3,870,000.00	$13,310,000.00	$41,430,000.00

	-$120,000.00	$670,000.00	$4,540,000.00	$17,850,000.00	$59,280,000.00

Valuation (NPV)	$47,356,978.10

Thank you

Appendix



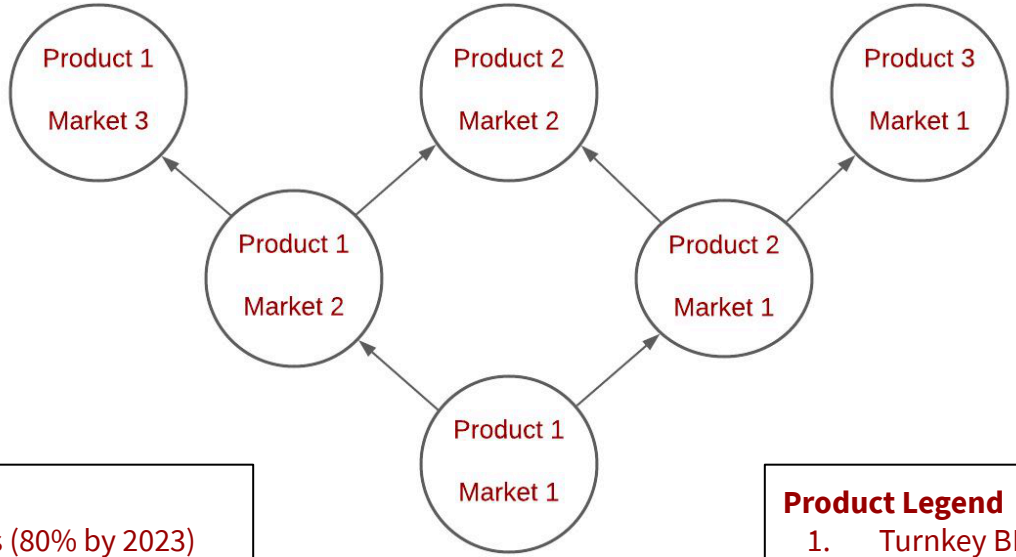
Introduction

Vision: create a safe, healthy, and productive indoor environment for all.

Mission: become a leading global provider of building management and automation systems.

Goal: develop intelligent building automation solutions enabled by IoT and that delight property owners and building operators, starting with the indoor farming segment.

Bowling Pin Diagram



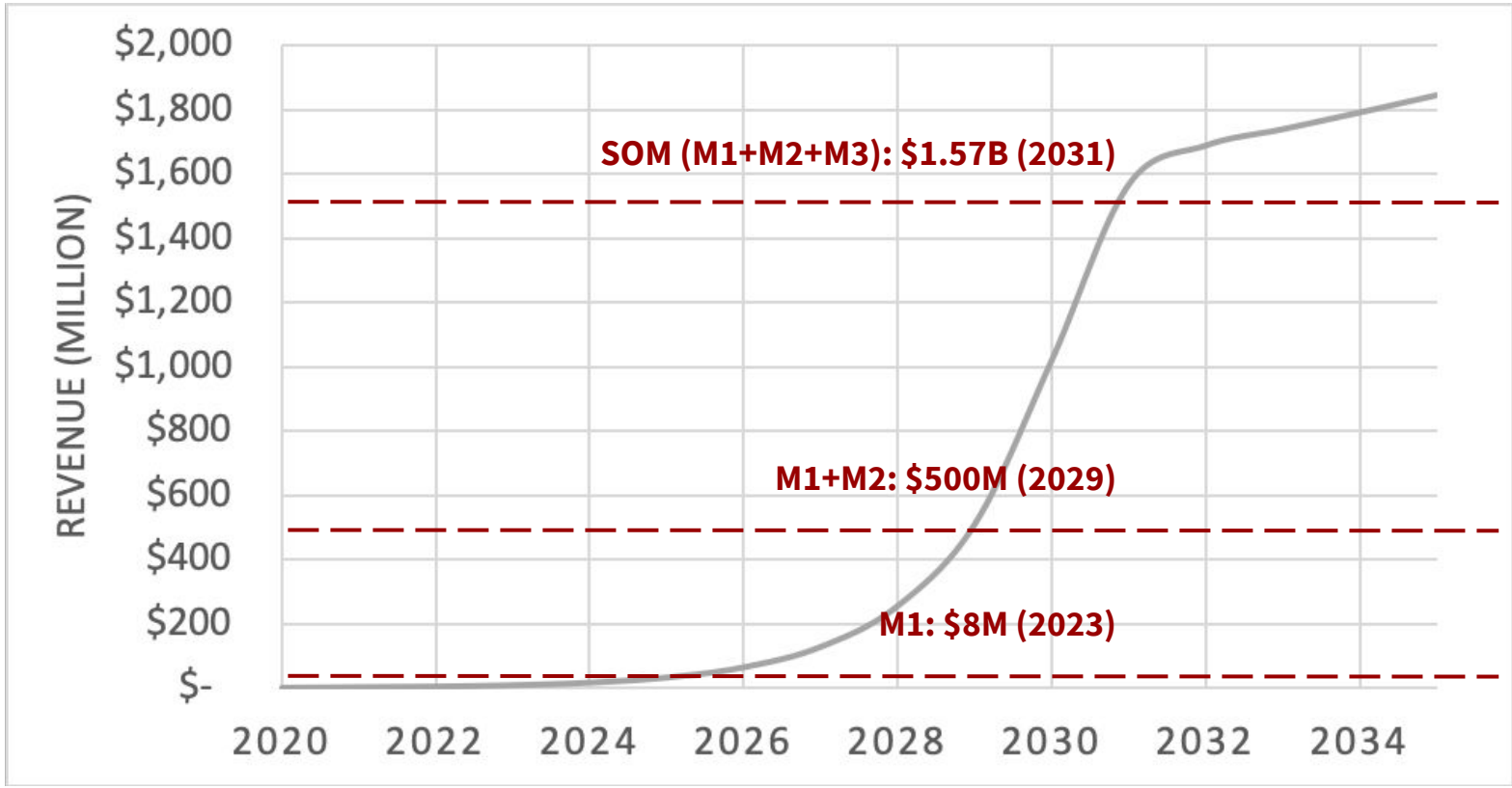
Market Legend

1. M1: SG Indoor Farms (80% by 2023)
2. M2: APAC C&I (50% by 2029)
3. M3: Europe C&I (50% by 2031)

Product Legend

1. Turnkey BMS (system integrator)
2. enabled BAS
3. enabled

3s Framework



MVP: Two Paths

Path	Original Equipment Manufacturer (OEM)	System Integrator
Activities	1. Design IoT device based on 2. Procure components 3. Assemble device with casing 4. Develop firmware 5. Install on-site	1. Procure controller and sensors 2. Install on-site 3. Connect with existing systems
Advantages	• Flexible customization • Financially sustainable	• Ready to use • Manufacturer support
Disadvantages	• Electronics design cost and complexity • Manufacturing and distribution costs	• Solutions are wired • No data analytics capabilities • Manufacturers are far away


Device Enclosure


Temp sensor

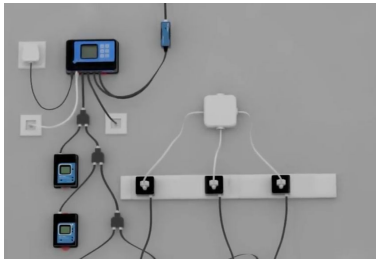

CO2 sensor


Micro-controller (“the platform”)


RH sensor



Transceiver


Connected controller and sensors


Controller


Sensors

Bill of Materials: IoT Sensors

Component Name	Part Number	Manufacturer	Description	Picture	Unit Price	Supplier	Quantity	Total
Board and microcontroller	Arduino Uno Rev3 SMD -- A000066	Arduino	Board with processor and attachment ports		$29.86	RS-Online	2	$59.72
Temperature Sensor	MAX31889ALT+	Maxim Integrated	Board Mount Temperature Sensors 0.1 degrees C accuracy, 1-Wire Temperature Sensor		$4.40	Mouser	2	$8.80
Relative Humidity Sensor	SHT31-DIS-F2.5kS	Sensirion	Board Mount Humidity Sensors RH/ T Sensor		$9.11	Mouser	2	$18.22
CO2 Sensor	T6713-6H	Amphenol Advanced Sensors	Air Quality Sensors CO2 Sensor Module for AAS-AQS-UNO		$122.54	Mouser	2	$245.08
Light Sensor	TSL2561T ams	ams	TSL2561T ams, Ambient Light Sensor Unit Surface Mount 6-Pin TMB		$4.41	RS-Online	2	$8.82
Transceiver	LR1110IMLTRT	Semtech	RF Transceiver LONG RANGE WI-FI LOC LORA TRANSCEIVER		$14.93	Mouser	2	$29.86
ADCs	AD7795BRUZ	Analog Devices	Analog Devices AD7795BRUZ, 16-Bit Serial ADC Differential Input, 24-Pin TSSOP		$16.02	RS-Online	8	$128.16
Device enclosure	NSYTBP342912T	Schneider Electric	Plastic Enclosure, Flame Retardant, Multipurpose, Polycarbonate, 341 mm, 291 mm, 128 mm, IP66		$106.51	Element 14	2	$213.02
							Total	**$720 for 2 sensors**

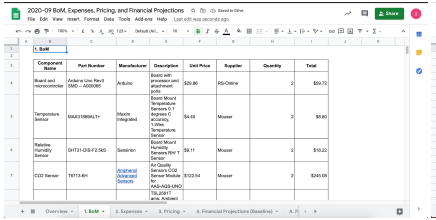
Detailed bill of materials for the OEM MVP path has been been made for IoT Sensors

MVP: Model

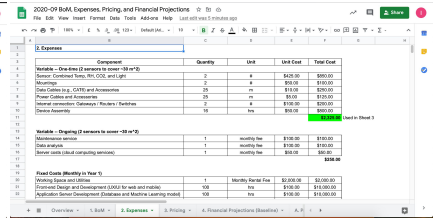
- System lays down the foundation for the prediction and optimization model
- Numerous options (depends on requirements)
- Potential: neural network (supervised learning)
- Consists of several “layers” of data -- potential structure:
  - Layer one: sensor measurements (i.e., raw sensor data)
  - Layer two: equipment operational logs
  - Layer three: output (yield, harvest time, crop quality)
- Use-cases:
  - Receive recommendations for climate controls (e.g., adjust temperature, increase humidity)
  - Predict crop output, harvest time, and quality
  - Automate climate controls

Financial Model Process

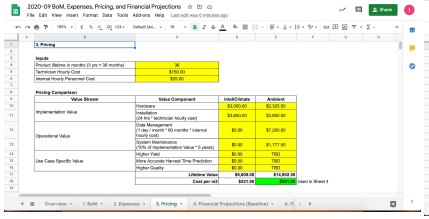
1. BoM



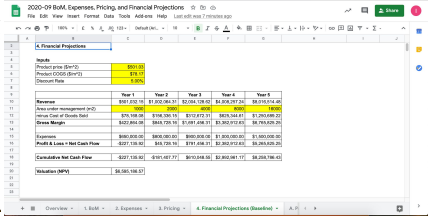
2. Expenses



3. Pricing



4. Financial Projections



Hardware Costs

Variable Expense

COGS and Price

NPV and